A First Course in Abstract Algebra

Section 18: Rings and Fields

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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18 Rings and Fields

Exercise 18.1: A product modulo 24

Multiply the integer representatives first: $12 \cdot 16 = 192$. Since $192 = 8 \cdot 24$, its remainder modulo 24 is 0. Thus the product in $\mathbb{Z}_{24}$ is $\boxed{0}$.

Exercise 18.2: A product modulo 32

The integer product is $16 \cdot 3 = 48 = 32 + 16$. Reduction modulo 32 therefore gives $\boxed{16}$.

Exercise 18.3: A negative product modulo 15

Compute $11(-4) = -44$. Adding the multiple 45 of 15 gives $-44 \equiv 1 \pmod{15}$. The product is $\boxed{1}$.

Exercise 18.4: A negative product modulo 26

We have $20(-8) = -160 = -7 \cdot 26 + 22$. Therefore the representative in $\{0, \ldots, 25\}$ is $\boxed{22}$.

Exercise 18.5: Multiplication in a product ring

Operations in $\mathbb{Z}_5 \times \mathbb{Z}_9$ are performed separately in each coordinate. Hence

$$(2, 3)(3, 5) = (6 \bmod 5, 15 \bmod 9) = \boxed{(1, 6)}.$$

Exercise 18.6: Two modular coordinates

In $\mathbb{Z}_4 \times \mathbb{Z}_{11}$ the product is

$$(-3, 5)(2, -4) = (-6, -20) = (2, 2),$$

because $-6 \equiv 2 \pmod{4}$ and $-20 \equiv 2 \pmod{11}$.

Exercise 18.7: The multiples of an integer

For $n \geq 1$, sums, negatives, and products of multiples of n remain multiples of n: $na - nb = n(a - b)$ and $(na)(nb) = n(nab)$. The associative and distributive laws are inherited from $\mathbb{Z}$. Thus $n\mathbb{Z}$ is a commutative ring. If it has unity u, applying $u \cdot n = n$ in the integers gives $u = 1$. Consequently it has unity exactly when $n = 1$, and it is never a field: when $n = 1$, the nonzero integer 2 has no inverse in $\mathbb{Z}$. If $n = 0$ is allowed, $0\mathbb{Z} = \{0\}$ is the zero ring, whose identity equals zero and which is not a field.

Exercise 18.8: Positive integers

Both addition and multiplication are closed on $\mathbb{Z}^+$. Nevertheless addition is not a group operation: 0 is absent and no positive integer has its additive inverse in the set. Thus these operations do not form a ring.

Exercise 18.9: The product of two integer rings

Addition and multiplication are componentwise, so each ring axiom follows in each coordinate from its counterpart in $\mathbb{Z}$. The result is a commutative ring with unity $(1,1)$. It is not a field: $(1,0)$ is nonzero but cannot have an inverse, because its product with any (a,b) has second coordinate zero.

Exercise 18.10: An even-integer coordinate

The subset $2\mathbb{Z}\times\mathbb{Z}$ is closed under subtraction and multiplication, so it is a commutative subring of $\mathbb{Z}\times\mathbb{Z}$. If (u,v) were its unity, multiplying $(2,0)$ would require $2u=2$, hence $u=1$, contrary to $u\in 2\mathbb{Z}$. It has no unity and is not a field.

Exercise 18.11: Integral coefficients with a square root

For integers a,b,c,d,

$$(a+b\sqrt{2})-(c+d\sqrt{2})=(a-c)+(b-d)\sqrt{2},\quad (a+b\sqrt{2})(c+d\sqrt{2})=(ac+2bd)+(ad+bc)\sqrt{2}.$$

Thus $\mathbb{Z}[\sqrt{2}]$ is a commutative subring of $\mathbb{R}$, with unity 1. It is not a field: an inverse of 2 would be $1/2$, which is not $a+b\sqrt{2}$ with integers a,b. Indeed $b\neq 0$ would make $\sqrt{2}$ rational, and $b=0$ would require $a=1/2$.

Exercise 18.12: Rational coefficients with a square root

The subtraction and multiplication formulas in Exercise 18.11 also hold for rational coefficients, proving that this set is a commutative ring with unity. For $a+b\sqrt{2}\neq 0$,

$$(a+b\sqrt{2})^{-1}=\frac{a-b\sqrt{2}}{a^2-2b^2}.$$

The denominator is nonzero: otherwise either $a=b=0$ or $\sqrt{2}=a/b$ would be rational. The displayed inverse again has rational coefficients. Therefore $\mathbb{Q}(\sqrt{2})$ is a field.

Exercise 18.13: Pure imaginary numbers

Addition is closed, but multiplication is not: $i\cdot i=-1$ is not purely imaginary. Since a ring requires a binary multiplication on its underlying set, this is not a ring under the indicated operations.

Exercise 18.14: Units in the integers

A unit $u\in\mathbb{Z}$ must satisfy $uv=1$ for an integer v. The only integer factors of 1 are 1 and -1. Thus the units are $\boxed{\{1,-1\}}$, each its own inverse.

Exercise 18.15: Units in a product of integer rings

The equation $(a,b)(c,d)=(1,1)$ is equivalent to $ac=1$ and $bd=1$. Therefore each of a,b must be 1 or -1, independently. The four units are $\boxed{(1,1),(1,-1),(-1,1),(-1,-1)}$.

Exercise 18.16: Units modulo five

Since 5 is prime, every nonzero residue is a unit. Explicitly, $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, and $4^{-1} = 4$, as multiplication verifies modulo 5. The units are $\boxed{1, 2, 3, 4}$.

Exercise 18.17: Units in the rational field

Every nonzero rational a/b has rational inverse b/a. Zero cannot multiply any rational to give 1. Thus the unit group is $\boxed{\mathbb{Q} \setminus \{0\}}$.

Exercise 18.18: Units in a three-factor ring

An inverse must work in all three coordinates. The first and last integer coordinates must therefore be ± 1, and the rational coordinate must be nonzero. All units are

$$\boxed{\{(\varepsilon, q, \delta) : \varepsilon, \delta \in \{1, -1\},\ q \in \mathbb{Q} \setminus \{0\}\}}.$$

The inverse is $(\varepsilon, q^{-1}, \delta)$.

Exercise 18.19: Units modulo four

The invertible residues are precisely those relatively prime to 4, namely $1, 3$. Directly, $1 \cdot 1 = 1$ and $3 \cdot 3 = 9 \equiv 1 \pmod 4$, whereas products with 0 or 2 cannot be odd. Thus the units are $\boxed{1, 3}$.

Exercise 18.20: The matrix ring over two elements

(a) Each of the four entries has two choices, giving $2^4 = \boxed{16}$ matrices. **(b)** A matrix over the field $\mathbb{Z}_2$ is invertible exactly when its determinant is nonzero. Its first column can be any of the three nonzero vectors; its second must be outside the span of the first, leaving two choices. The six units are

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Every listed determinant is 1 in $\mathbb{Z}_2$, and the counting argument shows that none are missing.

Exercise 18.21: A homomorphism that does not preserve unity

Use $\phi : \mathbb{Z} \longrightarrow \mathbb{Z} \times \mathbb{Z}$, $\phi(n) = (n, 0)$. Both addition and multiplication are preserved componentwise. However, $\phi(1) = (1, 0)$ is neither the zero $(0, 0)$ nor the target unity $(1, 1)$. The book's ring-homomorphism convention does not require unity preservation.

Exercise 18.22: The determinant map

For $n \geq 2$, determinants preserve multiplication but not addition. Take $A = \operatorname{diag}(1, 0, 0, \ldots)$ and $B = \operatorname{diag}(0, 1, 1, \ldots)$. Both determinants are zero, whereas $A + B = I_n$ has determinant 1. Hence $\det(A + B) \neq \det A + \det B$, and the map is not a ring homomorphism. For the exceptional case $n = 1$, the determinant is simply the entry, so it is an isomorphism $M_1(\mathbb{R}) \to \mathbb{R}$.

Exercise 18.23: Homomorphisms from the integers to themselves

Additivity forces $\phi(n) = n\phi(1)$ for every integer n, including negative integers. Set $k = \phi(1)$. Multiplicativity applied to $1^2 = 1$ gives $k^2 = k$, so $k = 0$ or 1. Both choices work. Thus the only maps are the zero map and the identity map.

Exercise 18.24: Homomorphisms into a product

Write $\phi(1) = (u, v)$. Additivity gives $\phi(n) = (nu, nv)$, and multiplicativity requires $(u^2, v^2) = (u, v)$. In $\mathbb{Z}$, an idempotent is 0 or 1. Consequently there are four homomorphisms:

$$n \mapsto (0,0), \quad n \mapsto (n,0), \quad n \mapsto (0,n), \quad n \mapsto (n,n).$$

Each preserves the two ring operations by direct substitution.

Exercise 18.25: Homomorphisms out of a product

Let $e_1 = (1,0)$ and $e_2 = (0,1)$. Since they are idempotent, their images u, v in $\mathbb{Z}$ lie in $\{0,1\}$. Since $e_1 e_2 = 0$, we must have $uv = 0$. Additivity then gives $\phi(a,b) = au + bv$. The possible pairs (u,v) are $(0,0), (1,0), (0,1)$, yielding exactly the zero map and the two coordinate projections.

Exercise 18.26: Three source coordinates

The three coordinate idempotents must map to 0 or 1 in $\mathbb{Z}$, and the product of any two different images must be zero. Therefore at most one image is 1. There are four choices: all zero, or one of the three coordinate positions equal to 1. These give the zero map and the three projections, so the answer is $\boxed{4}$.

Exercise 18.27: A false zero-product argument for matrices

The factorization is valid because I_3 commutes with X. The invalid step is inferring that a factor must be zero: matrix rings have zero divisors. For example,

$$X = \operatorname{diag}(1, 1, -1)$$

satisfies $X^2 = I_3$ while $X \neq I_3$ and $X \neq -I_3$. Here both $X - I_3$ and $X + I_3$ are nonzero, although their product is zero.

Exercise 18.28: A quadratic congruence modulo fourteen

Factor the polynomial as $(x+3)(x-2)$. A product is zero modulo 14 exactly when it is zero modulo both 2 and 7. Modulo 2 the original polynomial is $x(x+1)$, zero for both residues. Modulo 7, the two roots are 2 and $-3 = 4$. Lifting these residues to $\mathbb{Z}_{14}$ gives

$$\boxed{x = 2, 4, 9, 11}.$$

For example, at $x = 4$, neither $x + 3 = 7$ nor $x - 2 = 2$ is zero modulo 14, but their product is. This is precisely the zero-divisor issue in Exercise 18.27.

Exercise 18.29: Correcting the definition of a field

The missing requirement is commutativity of multiplication. A field is a *commutative* ring with unity $1 \neq 0$ such that every nonzero element has a multiplicative inverse; equivalently its nonzero elements form an abelian group under multiplication. The printed definition, without commutativity, describes a division ring.

Exercise 18.30: Correcting the definition of a unit

Magnitude is irrelevant in a general ring. In a ring with unity 1, an element u is a unit if there exists v in the ring with $uv = vu = 1$. Both equations are required in a possibly noncommutative ring.

Exercise 18.31: A product of nonzero elements that vanishes

In $\mathbb{Z}_6$, the residues 2 and 3 are nonzero, but $2 \cdot 3 = 6 = 0$. This supplies the requested example of nonzero zero divisors.

Exercise 18.32: A subring with its own unity

In $R = \mathbb{Z} \times \mathbb{Z}$, let $S = \mathbb{Z} \times \{0\}$. It is closed under subtraction and multiplication, so it is a subring. Its unity is $(1, 0)$, which is nonzero and differs from the unity $(1, 1)$ of R. Indeed $(1, 0)(a, 0) = (a, 0) = (a, 0)(1, 0)$.

Exercise 18.33: True or false: ring and field axioms

(a) True: a field is a ring with additional properties.

(b) False: $2\mathbb{Z}$ has no multiplicative identity.

(c) False: $\mathbb{Z}_2$ has just one unit, 1, since $-1 = 1$ there.

(d) False: $\mathbb{Z}_5$ has four units.

(e) True: $\mathbb{Z} \subset \mathbb{Q}$ is a subring but not a subfield, because $2^{-1} = 1/2 \notin \mathbb{Z}$.

(f) False: distributivity is an essential ring axiom, linking the two operations.

(g) True: commutative multiplication is part of the definition of a field.

(h) True: this is the nonzero multiplicative group in the field definition.

(i) True: the additive group of a ring is required to be abelian.

(j) True: the additive-group axiom provides an additive inverse for every element.

Exercise 18.34: Pointwise multiplication of functions

The operations are pointwise: $(fg)(x) = f(x)g(x)$ and $(f + g)(x) = f(x) + g(x)$. For every point x,

$$((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x).$$

Thus multiplication is associative. Likewise

$$(f(g+h))(x) = f(x)(g(x)+h(x)) = f(x)g(x) + f(x)h(x) = (fg+fh)(x),$$

and $((f+g)h)(x) = (fh+gh)(x)$. Equality at every input means equality of functions, proving the required associative and both distributive axioms.

Exercise 18.35: Evaluation preserves multiplication

For the evaluation map $\phi_a(f) = f(a)$, the definition of pointwise product gives

$$\phi_a(fg) = (fg)(a) = f(a)g(a) = \phi_a(f)\phi_a(g).$$

This verifies the multiplicative requirement, without assuming that a general function map preserves products.

Exercise 18.36: Ring isomorphism is an equivalence relation

Reflexivity: the identity map preserves both operations and is bijective. **Symmetry:** if $\phi : R \to S$ is an isomorphism, write $u = \phi(a)$, $v = \phi(b)$. Then $\phi^{-1}(u+v) = a+b$ and $\phi^{-1}(uv) = ab$, so its inverse is also an isomorphism. **Transitivity:** if $\psi : S \to T$ is an isomorphism as well, then

$$(\psi\phi)(a+b) = \psi(\phi(a)+\phi(b)) = (\psi\phi)(a) + (\psi\phi)(b),$$

and the same calculation with products proves multiplicativity. The composition is bijective. These are exactly the three equivalence-relation properties.

Exercise 18.37: The group of units

Let u, v be units. Their product is a unit because

$$(uv)(v^{-1}u^{-1}) = 1 = (v^{-1}u^{-1})(uv).$$

Thus multiplication is closed on the units; its associativity is inherited from the ring. The unity is a unit, and the inverse of a unit is itself a unit, with inverse u. Consequently all group axioms hold. The order reversal in $(uv)^{-1}$ matters when multiplication is not commutative.

Exercise 18.38: A difference-of-squares criterion

If R is commutative, expansion gives $(a+b)(a-b) = a^2 - ab + ba - b^2 = a^2 - b^2$. Conversely, suppose the stated identity holds for every a, b. Expanding its right side and canceling a^2 and $-b^2$ in the additive group gives $-ab + ba = 0$, hence $ab = ba$. Therefore multiplication is commutative.

Exercise 18.39: Zero multiplication

Define $ab = 0$ for every pair. This is a closed multiplication, and both $(ab)c$ and $a(bc)$ are zero. Furthermore $a(b+c) = 0 = 0+0 = ab+ac$ and $(a+b)c = 0 = ac+bc$. Together with the assumed abelian additive group, these facts prove all ring axioms.

Exercise 18.40: Two pairs of nonisomorphic rings

The rings $2\mathbb{Z}$ and $3\mathbb{Z}$. An additive isomorphism must carry the additive generator 2 to either 3 or -3. If $\phi(2) = 3\varepsilon$ with $\varepsilon = \pm 1$, additivity gives $\phi(4) = 6\varepsilon$, whereas multiplicativity gives $\phi(4) = \phi(2)^2 = 9$. These cannot agree.

The fields $\mathbb{R}$ and $\mathbb{C}$. Any field isomorphism carries unity to unity: its image of 1 acts as an identity on the whole target. Therefore a supposed map $\phi : \mathbb{C} \to \mathbb{R}$ would give $\phi(i)^2 = \phi(-1) = -1$. No real number has square -1. Hence these fields cannot be isomorphic.

Exercise 18.41: The prime-characteristic binomial identity

In a commutative ring the binomial expansion is valid. For $1 \le k \le p-1$, the binomial coefficient $\binom{p}{k}$ is divisible by the prime p: its denominator $k!(p-k)!$ has no factor p, whereas its numerator $p!$ has one. In $\mathbb{Z}_p$ all these intermediate coefficients vanish. Thus

$$(a+b)^p = a^p + \sum_{k=1}^{p-1} \binom{p}{k} a^{p-k} b^k + b^p = a^p + b^p.$$

Exercise 18.42: The unity of a subfield

Let e be the unity of a subfield $K \subset F$. It is nonzero, and $e^2 = e$ in F. Multiply this equation by the inverse of e in the field F to obtain $e = 1_F$. Cancellation in a field is why a subfield cannot exhibit the different-unity phenomenon of Exercise 18.32.

Exercise 18.43: Uniqueness of a multiplicative inverse

If b and c are both two-sided inverses of a, then

$$b = b1 = b(ac) = (ba)c = 1c = c.$$

Associativity and the two inverse equations justify every step. Thus a unit has exactly one two-sided inverse.

Exercise 18.44: Idempotents and a product ring

(a) If $a^2 = a$, $b^2 = b$, and multiplication is commutative, then $(ab)^2 = a^2 b^2 = ab$. Hence the product is idempotent.

(b) An idempotent residue x satisfies $x(x-1) \equiv 0$. In $\mathbb{Z}_6$, testing the six residues gives $\{0, 1, 3, 4\}$. In $\mathbb{Z}_{12}$, the requirement is that $x \equiv 0$ or 1 modulo each of 4 and 3, giving $\{0, 1, 4, 9\}$. Idempotence is coordinatewise, so all sixteen idempotents are

$$\boxed{\{0,1,3,4\} \times \{0,1,4,9\}}.$$

Exercise 18.45: An idempotent projection matrix

Put $B = (A^TA)^{-1}$, so $P = ABA^T$. Associativity of matrix multiplication, with no illicit reordering, gives

$$P^2 = ABA^TABA^T = AB(A^TA)BA^T = ABI_nA^T = ABA^T = P.$$

The dimensions are consistent: A is $m \times n$, B is $n \times n$, and P is $m \times m$. Thus it is an idempotent in $M_m(\mathbb{R})$. The printed reference to $n \times n$ matrices is a dimension slip: the formula for P gives $m \times m$.

Exercise 18.46: Adding nilpotent elements

Choose positive integers r, s with $a^r = 0$ and $b^s = 0$. Because the ring is commutative, expand $(a+b)^{r+s-1}$. In each term $a^kb^{r+s-1-k}$, either $k \geq r$, or else $k \leq r-1$ and $r+s-1-k \geq s$. The term is zero in either case. The entire expansion vanishes, so $a+b$ is nilpotent.

Exercise 18.47: Testing nilpotence using squares

If no nonzero nilpotent exists, an element satisfying $x^2 = 0$ must be zero. Conversely, suppose every square-zero element is zero. If a were nonzero nilpotent, choose the least $n \geq 2$ with $a^n = 0$. Then $b = a^{n-1} \neq 0$ by minimality, but $b^2 = a^{2n-2} = 0$ because $2n-2 \geq n$. This contradicts the square-zero condition.

Exercise 18.48: The subring test

A subring necessarily contains zero and is closed under subtraction and multiplication. Conversely suppose the three listed conditions hold. Setting $a = 0$ shows $-b \in S$; then $a + b = a - (-b) \in S$. Thus S is an additive subgroup of the abelian group $(R, +)$. Closure supplies a multiplication on S, and its associativity and both distributive laws are inherited from R. Hence S is a ring with the induced operations. A shared multiplicative unity is not required by the book's subring convention.

Exercise 18.49: Intersections of subrings and subfields

(a) Every subring in the family contains 0. If a, b are in their intersection, then $a - b$ and ab lie in every member, hence in the intersection. Apply Exercise 18.48.

(b) For a nonempty family of subfields of F, their common unity is 1_F by Exercise 18.42. The intersection is a subring and contains this nonzero unity. If $a \neq 0$ belongs to it, its unique inverse in F lies in every subfield, hence in the intersection. It is therefore a field. With the convention that an empty intersection is the ambient object, the empty-family conclusions hold as well.

Exercise 18.50: A one-sided annihilator is a subring

For $I_a = \{x : ax = 0\}$, first $a0 = 0$, so $0 \in I_a$. If $x, y \in I_a$, distributivity gives $a(x-y) = ax - ay = 0$. Associativity gives $a(xy) = (ax)y = 0y = 0$. Thus the subring test applies. No commutativity of the ambient ring is needed.

Exercise 18.51: The subring generated by one element

Let H consist of all finite sums $\sum_{k=1}^{N} m_k a^k$ with integer coefficients. This is exactly the additive subgroup generated by the positive powers of a. It contains a and is closed under subtraction. Products remain in H, because distributivity expresses a product as a finite integer combination of powers a^{i+j}. Therefore H is a subring containing a, so $R_a \subseteq H$ by the definition of R_a as an intersection. Conversely every subring containing a contains all positive powers and their integer linear combinations, so $H \subseteq R_a$. The two sets are equal. A constant term is not included: subrings here need not contain the ambient unity.

Exercise 18.52: Two simultaneous congruences

For coprime r, s, the map $\mathbb{Z}_{rs} \to \mathbb{Z}_r \times \mathbb{Z}_s$, $[x]_{rs} \mapsto ([x]_r, [x]_s)$, is a ring isomorphism as in the text. In particular it is onto, so the pair $([m]_r, [n]_s)$ has a preimage. Any integer representative x of that preimage satisfies both required congruences. Concretely, choose integers u, v with $ur + vs = 1$; then $x = nur + mvs$ has residues m modulo r and n modulo s.

Exercise 18.53: The general Chinese remainder theorem

(a) For pairwise coprime positive moduli $b_1, \ldots, b_n$, put $B = \prod b_i$. The map

$$\Phi : \mathbb{Z}_B \longrightarrow \prod_{i=1}^{n} \mathbb{Z}_{b_i}, \qquad [x]_B \mapsto ([x]_{b_1}, \ldots, [x]_{b_n})$$

is a well-defined ring homomorphism. Its kernel is zero: if every b_i divides x, their pairwise coprimality implies $B \mid x$. Both finite sets have B elements, so injectivity implies bijectivity. This proves the direct-product isomorphism.

(b) Surjectivity applied to the tuple of residues a_i produces the desired integer solution. Adding a sufficiently large multiple of B makes it positive without changing any congruence. For an explicit formula, let $B_i = B/b_i$ and choose u_i with $u_i B_i \equiv 1 \pmod{b_i}$. Then $x = \sum_i a_i u_i B_i$ has residue a_j modulo b_j, because every other summand is divisible by b_j.

Exercise 18.54: Recovering commutative addition in a division ring

We must prove the missing additive commutativity. Distributivity and additive cancellation first show $a0 = 0 = 0a$, since, for example, $a0 = a(0 + 0) = a0 + a0$. The multiplicative identity 1 on the nonzero elements also satisfies $10 = 01 = 0$, so it acts as unity on all of S. Expand the same product in two ways:

$$(1 + 1)(a + b) = a + b + a + b = (1 + 1)a + (1 + 1)b = a + a + b + b.$$

Cancel the initial a and final b in the additive group to obtain $b + a = a + b$. Thus $(S, +)$ is abelian. Multiplication is associative when all factors are nonzero by the assumed group axiom; if any factor is zero both associated products are zero. The assumed distributive laws now give all ring axioms, with nonzero unity and inverses for every nonzero element. Hence S is a division ring.

Exercise 18.55: Boolean rings are commutative

For any a, idempotence of $-a$ gives $(-a)^2 = -a$. But $(-a)^2 = a^2 = a$, so $a = -a$, equivalently $2a = 0$. Next expand the identity $(a+b)^2 = a+b$:

$$a^2 + ab + ba + b^2 = a + b.$$

Since $a^2 = a$ and $b^2 = b$, cancellation yields $ab + ba = 0$. Therefore $ab = -ba = ba$, using the already proved fact that every element equals its negative. The ring is commutative.

Exercise 18.56: A Boolean ring of subsets

(a) Abbreviate $\varnothing, \{a\}, \{b\}, \{a,b\}$ by $0, A, B, S$. The addition is symmetric difference and multiplication is intersection:

$+$	0	A	B	S
0	0	A	B	S
A	A	0	S	B
B	B	S	0	A
S	S	B	A	0

$\cdot$	0	A	B	S
0	0	0	0	0
A	0	A	0	A
B	0	0	B	B
S	0	A	B	S

.

(b) Associate to each subset $T \subseteq S$ its indicator function $\chi_T : S \to \mathbb{Z}_2$. At each point, $\chi_{T \triangle U} = \chi_T + \chi_U$ and $\chi_{T \cap U} = \chi_T \chi_U$, with operations in $\mathbb{Z}_2$. The bijection between subsets and their indicator functions therefore transfers the pointwise ring axioms to these set operations. The additive identity is $\varnothing$, each subset is its own additive inverse, and the unity is S. Finally $T \cap T = T$, so every element is idempotent and this is a Boolean ring.